

PROFESSIONAL SUMMARY

Ph.D. candidate in Physical Chemistry (computational; expected May 2027) with four years of experience modeling matter across scales—electronic structure (DFT), atomistic molecular dynamics, and mesoscale coarse-grained simulation—for systems spanning inorganic nanomaterials, metal–organic frameworks, soft matter, and biomolecules. Develops and validates simulation models from scratch, including force-field parameterization and statistical benchmarking against higher-resolution and experimental data, and scales production workflows across national HPC systems. Translates simulation results into design rules for experimental collaborators.

EDUCATION

Johns Hopkins University, Baltimore, MD*Ph.D. in Chemistry (Computational Chemistry)*

Aug 2022 – May 2027

M.A. in Chemistry

Aug 2025

University of Minnesota, Twin Cities, Minneapolis, MN*B.A. in Chemistry, Minor in Mathematics, Dean's List (4 terms)*

Aug 2018 – Dec 2021

TECHNICAL SKILLS

Molecular Simulation: All-atom and coarse-grained Molecular Dynamics, force-field parameterization, enhanced sampling, performance optimization, GROMACS, NAMD, LAMMPS

Electronic Structure: DFT, periodic and cluster calculations, Quantum ESPRESSO, ORCA

Analysis & Programming: Python, SQL, NumPy, SciPy, pandas, MDAnalysis, scikit-learn, MATLAB, Bash, Git, statistical model validation, trajectory analysis, VMD, ChimeraX, OVITO, VESTA, Blender, machine learning

HPC & Workflows: SLURM, GPU/MPI configuration, multi-cluster job management (ACCESS: Bridges-2, Expanse, Delta), reproducible pipelines, LaTeX

Experimental Biology: Cell culture, transfection, PCR/qPCR, RNA extraction, gel electrophoresis, NanoDrop, microscopy

Languages: English (fluent), Mandarin Chinese (native)

RESEARCH EXPERIENCE

Johns Hopkins University, Baltimore, MD

Aug 2022 – Present

Graduate Researcher, Multiscale Simulation of Nano – Bio Interfaces, Hernandez Group (NSF Center for Sustainable Nanotechnology)

Model Development & Parameterization

- Developed a mesoscale DPD lipid model from MARTINI bead mappings and bonded terms, extending vesicle simulations to 60 nm systems and microsecond-scale dynamics inaccessible to all-atom methods.
- Built DPD force-field parameters for anionic and cationic thiol ligands and corrected Au–S Morse potentials, enabling stable all-atom simulation of charged nanoparticles at membrane interfaces.
- Performed periodic and cluster DFT calculations with Quantum ESPRESSO and ORCA to quantify dopant effects on the electronic structure of metal–organic frameworks.

Validation & Root-Cause Analysis

- Validated coarse-grained models against all-atom simulations and neutron spin-echo experiments using Kolmogorov–Smirnov statistics, van Hove functions, and Zilman–Granek fitting, achieving sub-nanometer structural agreement with all-atom reference and a 2,700-fold reduction in computational cost.
- Developed an extended-sampling protocol to eliminate finite-window artifacts in membrane relaxation-dynamics fitting, achieving quantitative agreement across the full accessible wavevector range.

Analysis Pipelines, HPC & Machine Learning

- Built trajectory-analysis pipelines with MDAAnalysis, NumPy, and SciPy for residue-contact, orientation, and conformational analysis, resolving the electrostatic determinants of protein adsorption on charged nanoparticle surfaces.
- Built SLURM job-orchestration and GPU/MPI configuration scripts for production simulations across four HPC systems, and led the group's NSF ACCESS supercomputing allocation renewal.
- Designed and trained gradient-boosted regression models with scikit-learn, XGBoost, and RDKit to predict aqueous solubility from molecular structure, and benchmarked scaffold-based against random data splitting, exposing 41–195% overestimation of generalization performance under the naive approach.

University of Minnesota, Minneapolis, MN

Oct 2020 – Jan 2022

Undergraduate Research Assistant (Inst. on the Biology of Aging & Metabolism)

- Ran cell-culture, transfection, and drug-treatment workflows for non-homologous end-joining (NHEJ) DNA-repair assays supporting senotherapeutic screening.
- Extracted RNA from mouse tissues and performed qPCR to evaluate a candidate anti-aging compound.
- Optimized assay conditions and quantified SIRT6 self-mono-ADP-ribosylation activity and contributed to four co-authored papers and preprints.

WORK EXPERIENCE

Center for Sustainable Nanotechnology (NSF-funded)

Jun 2024 – Jun 2025

Executive Committee Liaison

Elected by 35 graduate students and postdocs across 11 institutions—including Johns Hopkins, Princeton, Boston University, UIUC, and Brookhaven National Lab—to represent trainees in executive-committee meetings with center leadership. Organized and moderated recurring student/postdoc board meetings and relayed trainee concerns to management.

Summer Student Mentor

Mentored summer students through an introductory project on protein simulation—system setup, running MD, and basic trajectory analysis.

Johns Hopkins University, Department of Chemistry

Aug 2022 – May 2027

Graduate Teaching Assistant

Taught Physical Chemistry and General Chemistry lecture and lab courses(8 semesters, 2022–2026)

Peer Reviewer, ACS Materials Au, Nano Select, Scientific Reports

2024 – 2026

CONFERENCES

- American Chemical Society National Meeting, Poster, 2025
- Center for Sustainable Nanotechnology Semi-Annual Meetings, Poster (5 meetings, 2023–2025)

PUBLICATIONS

- **W. Xu**, X. Wei, G. Chong, U. Dahal, Q. Cui, R. Hernandez. Structural and Dynamical Consistency in Lipid Vesicles across All-Atom, MARTINI, and Dissipative Particle Dynamics. *RSC Advances* 2026
- **W. Xu**, J. Lim, R. Hernandez. Drivers to Preferential Orientational Binding of Proteins to Gold Nanoparticles. In preparation
- X. Wei, **W. Xu**, K. Mendis, Z. Rosenzweig, R. Hernandez. Binding Affinity between Polymer Dots (Pdots) and Ovalbumin Protein at Varying pH. Submitted to *Langmuir* 2026; arXiv:2608.22871
- L. Zhang, C. Soto-Palma, ..., **W. Xu**, et al. Atypical NF-κB Activation by Endogenous DNA Damage Drives Senescence and Aging. *Nature Communications*, under review 2026
- L.J. Zhang, R. Salekeen, C. Soto-Palma, ..., **W. Xu**, et al. Polyunsaturated Lipid Senolytics Exploit a Ferroptotic Vulnerability in Senescent Cells. *Cell Press Blue* 2026
- L.J. Zhang, R. Salekeen, C. Soto-Palma, ..., **W. Xu**, et al. Novel Flavonoid Senotherapeutics Identified by Phenotypic Drug Discovery Reduce Senescence and Improve Multiple Markers of Healthspan. *Aging Cell*, accepted 2026
- P. Robbins, L. Zhang, O. Elsallabi, ..., **W. Xu**, et al. Fucoidans Are Senotherapeutics that Enhance SIRT6-Dependent DNA Repair. *Research Square* 2025
- C. Deng, Y. Wang, C. Castillo, Y. Zhao, **W. Xu**, et al. Nanoscale Iron (Fe₃O₄) Surface Charge Controls Fusarium Suppression and Nutrient Accumulation in Tomato. *ACS Sustainable Chemistry & Engineering* 2024